

CODEALPHA INTERNSHIP PROGRAM

INTERNET OF THINGS(IoT)

TASK 4: IoT PROTOTYPE WITH ARDUINO

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Internship Domain: Internet of Things (IoT)

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CERTIFICATE OF COMPLETION

This is to certify that the case study titled "IoT and Artificial Intelligence Integration" has been completed as part of the academic requirements. The study explores of the project "**Temperature and humidity monitoring system**". The project demonstrates the application of the **Internet of Things (IoT)** by using an Arduino microcontroller and a humidity sensor to monitor environmental humidity levels in real time. The sensor collects humidity data, which is processed by the Arduino and can be displayed or transmitted for further monitoring and analysis.

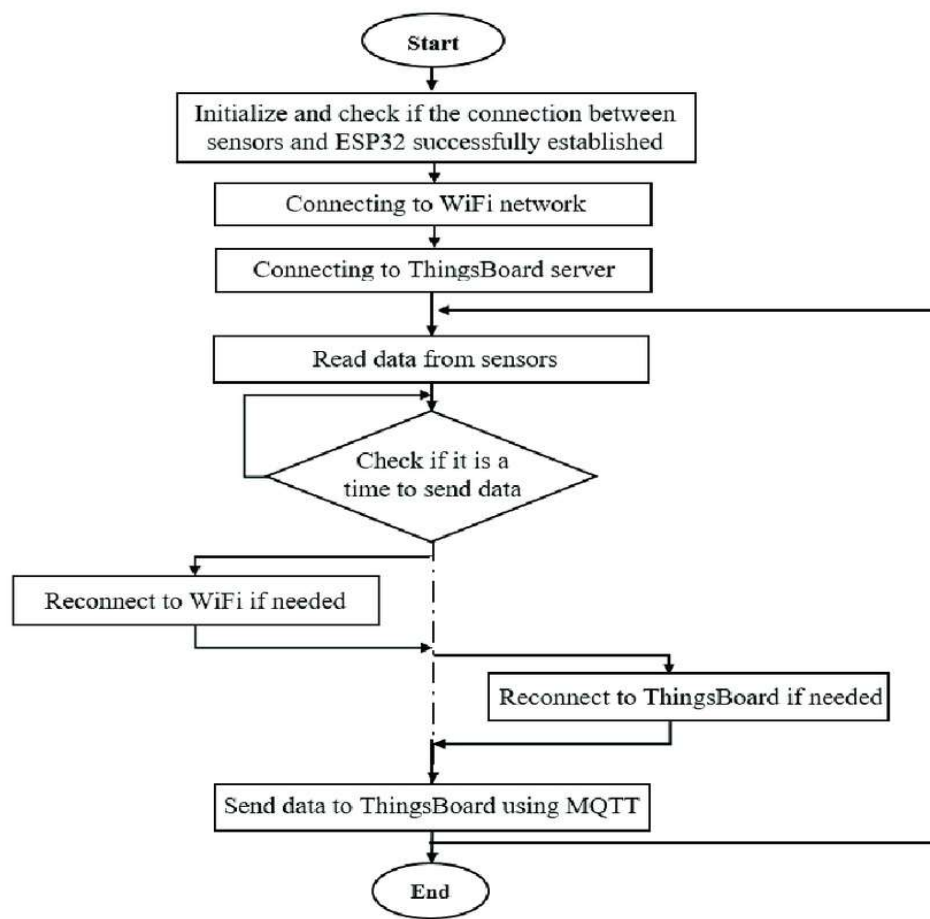
This project highlights the practical implementation of IoT concepts, including sensor interfacing, data acquisition, and environmental monitoring. It enhances knowledge of embedded systems, Arduino programming, and IoT-based automation, providing valuable hands-on experience in developing smart monitoring solutions.

PROJECT TITLE: TEMPERATURE AND HUMIDITY MONITORING SYSTEM

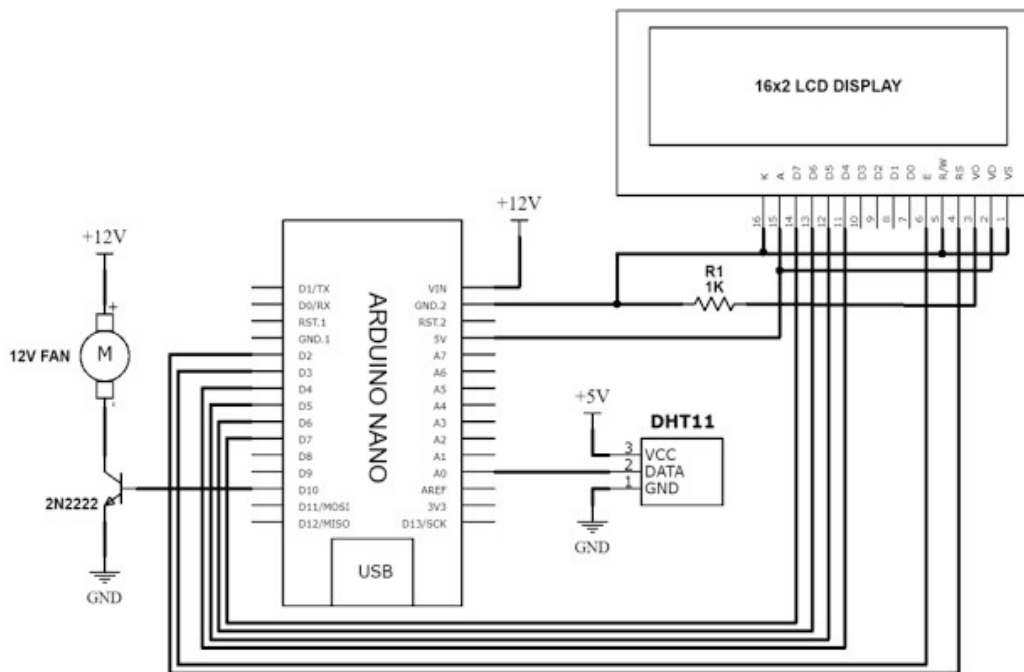
This project presents an IoT-based temperature and humidity monitoring system using an ESP32 microcontroller and a DHT22 sensor. The sensor measures environmental conditions and the Arduino processes the data and connects to a Wi-Fi network. The collected information is transmitted to the Things Board cloud platform using the MQTT protocol. The cloud platform displays the data on dashboards, enabling real-time remote monitoring through the internet. This system demonstrates a simple and cost-effective solution for environmental monitoring in applications such as smart homes, laboratories, warehouses, and industrial environments.

Key Words: IoT, Arduino nano, DHT22, MQTT, Things Board, Environmental Monitoring.

FLOW CHART:



CIRCUIT DIAGRAM:



CODE:

```
#include <DHT.h>

#define DHTPIN 2

#define DHTTYPE DHT11

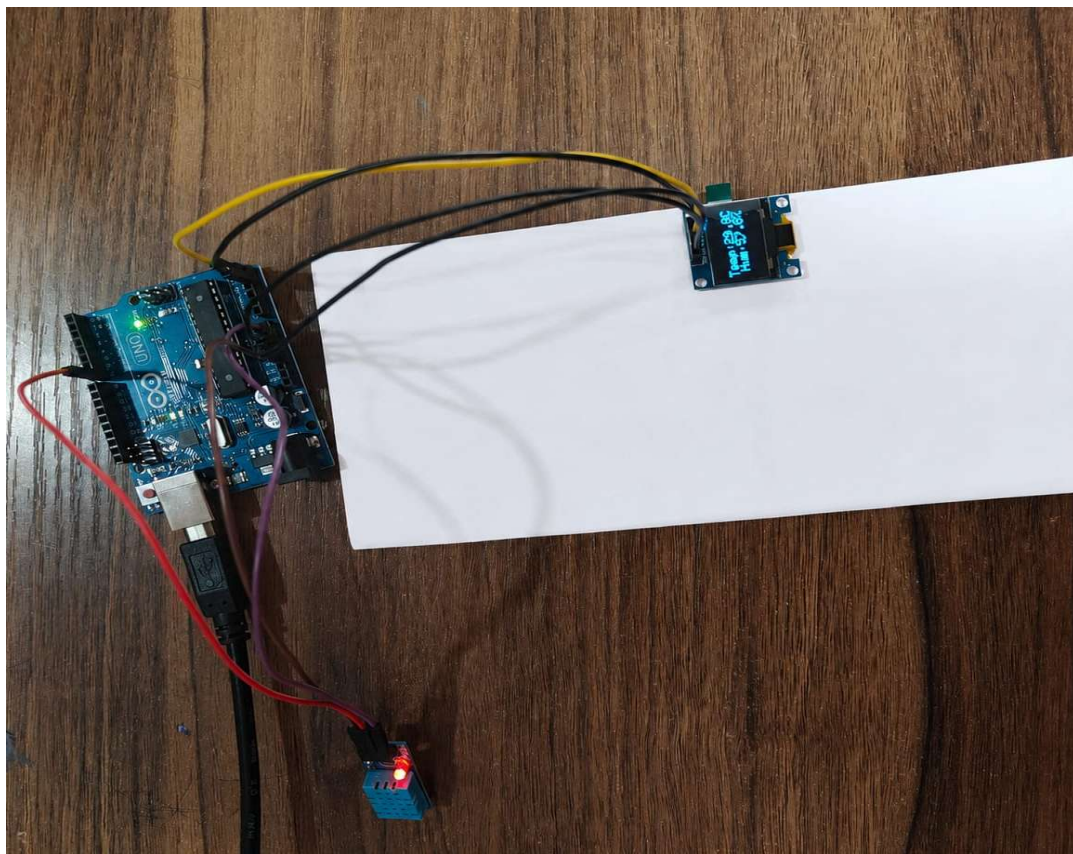
DHT dht(DHTPIN, DHTTYPE);

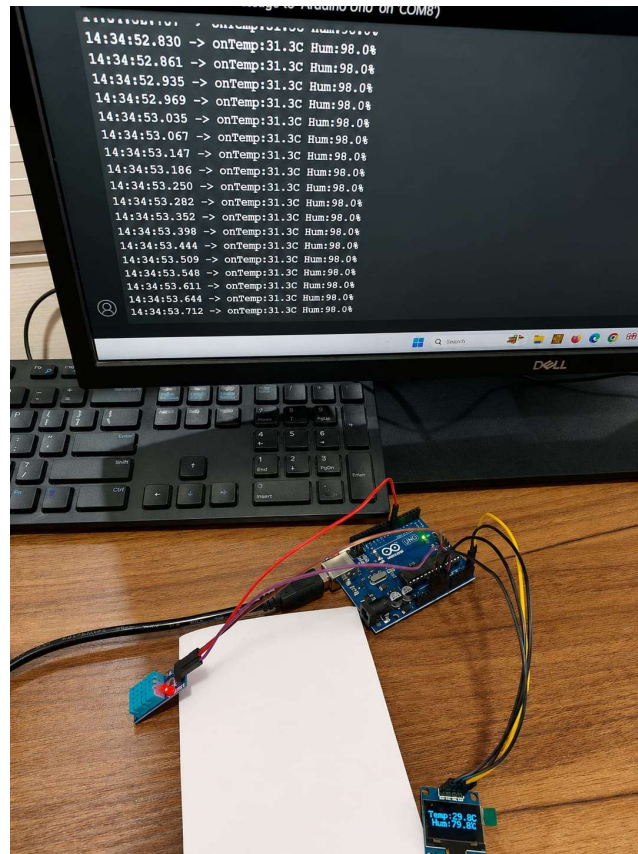
void setup() {
  Serial.begin(9600);
  dht.begin();
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();
  if (isnan(humidity) || isnan(temperature)) {
```

```
Serial.println("Failed to read from DHT11 sensor!");  
  
    return;  
}  
  
Serial.print("Temperature: ");  
Serial.print(temperature);  
Serial.print(" °C");  
Serial.print(" Humidity: ");  
Serial.print(humidity);  
Serial.println(" %");  
delay(2000);  
}
```

OUTPUT PICTURES:





ADVANTAGES AND APPLICATIONS:

Advantages:

- Real-time monitoring of temperature and humidity through the internet
- Wireless data transmission using Wi-Fi (no complex wiring required)
- Low-cost and energy-efficient system
- Supports automation and smart monitoring applications

Applications:

- Smart homes for Indoor Climate monitoring
- Laboratories for Maintaining controlled conditions
- Warehouse for storage environment monitoring
- Weathering monitoring stations

CONCLUSION

The **Temperature and Humidity Monitoring System using Arduino** successfully measures and displays real-time temperature and humidity values using the DHT11 sensor. This project demonstrates the practical application of IoT and embedded systems for environmental monitoring. It is simple, cost-effective, and reliable, making it suitable for homes, offices, greenhouses, laboratories, and industrial environments. The project also enhances understanding of sensor interfacing, Arduino programming, and real-time data monitoring, providing a strong foundation for developing advanced IoT-based smart monitoring systems.